Ram Calculus and Quintic Liénard Systems

M278: Half-Arc Matching, Multiplier Identities,
and Quantitative Perturbation Certificates

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M278

October 2026
Confidential research report; family 143, applicability L4

Abstract

We examine the pinned manuscript claiming at most two geometrically distinct isolated periodic orbits for every classical Liénard system with polynomial primitive of degree at most five, including arbitrary even coefficients. The complete half-arc, quadratic-fit, model-inequality, transport, coefficient partition, and sharpness argument is distinguished from the bounded results proved here. The actual private return-operator and perturbative-threshold passages supply a certification contract, not a global counting theorem. We derive an exact matching-derivative formula for the Floquet multiplier and a positive local factor relating matching displacement to return displacement, preserving isolated-root multiplicity. We also give exact rational sign and derivative margins for the source's two-cycle example, turning unspecified smallness into a conditional explicit remainder budget. An additional weak-shape comparison and exact positive-ray identity give a restricted nonodd degree-seven extension conditional on the source's global comparison package, with a two-cycle small-parameter family. These results retain the first-return, transverse-section, compact-domain, and uniform derivative hypotheses needed to count cycles, and do not exclude distant cycles or certify multiple roots by simple-root tests. Primary-public-source checks, finite exact tests, and bounded review evidence identify the scope of the conclusions. Contribution assessment: C2/5, source-specific exact interfaces and perturbation margins without an independent global theorem or established historical novelty.

Contents

1 Exact Target and Claim Discipline **3**

2 Direct Corpus Inputs and Public Context **3**
2.1 Public literature and version check 4

3 Half-Arcs and the Full Matching Domain **4**
3.1 Coordinates and exact transport 4
3.2 Why height zero does not require smoothness of the profile 5
3.3 The global geometric correspondence 5

4 Quadratic Fitting and the Central Model Estimate **6**
4.1 The exact global fit asserted by the source 6
4.2 Transport identities and derivative conventions 7
4.3 Riccati and Schwarzian structure of the source proof 7

5 Shape Transport and the Four Coefficient Cases **9**
5.1 Positive-height comparisons and boundary care 9
5.2 The unrestricted even part is retained 10

6 The Source's Sharpness Calculation **11**

7 Exact Matching and Multiplier Interfaces **11**

8 Exact Perturbation Budgets for the Sharpness Family **12**

9 A Conditional Shape Extension **14**

10 Benefits, Limitations, and Research Directions **15**

A Reproducibility and Bounded Review **16**

1 Exact Target and Claim Discipline

Let $F \in \mathbb{R}[x]$ have degree at most five and consider on $\mathbb{R}^2$

$$\dot{x} = y - F(x), \qquad \dot{y} = -x. \tag{1.1}$$

The damping polynomial in the equivalent equation $x'' + F'(x)x' + x = 0$ has degree at most four; the restoring force is exactly x. A limit cycle is the image of a nonconstant periodic solution isolated among periodic orbits. Geometric images, not multiplicities, are counted. A center's periodic annulus contributes zero limit cycles. The quantifier is over all real coefficients of F, all finite amplitudes, and all isolated cycles, including nonhyperbolic ones.

Source claim. The pinned OpenAI manuscript *Two limit cycles for quintic Liénard systems* [1], dated September 24, 2026, claims that the maximum is exactly two. It is the degree-five classical Liénard problem, not a bound for all degree-five planar polynomial vector fields or for arbitrary quintic restoring forces. The input pin is

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

All eight section files, `paper.tex`, and `figures/half-arcs.tex` were read directly. The local snapshot is an export; its identity is checked by public-byte comparison, not by assuming that an exported directory has Git metadata.

Category	Meaning here
Source claim	The global fit, model inequality, shape comparison, and complete two-cycle upper bound are the source's assertions.
Imported method	Smooth ODE dependence, the implicit function theorem, Liouville's determinant formula, variation of constants, and real one-variable root counting are classical.
Local result	The exact multiplier and displacement interfaces and the rational perturbation margins are proved explicitly below.
Conditional bridge	A specified C^1 remainder bound and certified return domain turn the exact margins into a numerical smallness threshold.
Not established	No independent global upper bound, new degree-only theorem, historical novelty, or computational speedup is asserted.

L4 measures applicability of the private corpus to the matching and counting interfaces. C2 measures this report's bounded contribution; neither rating certifies the source's headline theorem.

2 Direct Corpus Inputs and Public Context

The actual private source is `TR-item-015-astra.tex`, at its catalogued local path. Lines 250–361 give the return-operator theorem, its coverage and one-intersection requirements, complementary perturbation certificates, and the singular-chart uniformity barrier. Lines 486–577 give return-event differentiation, scalar divergence control of multipliers, and the explicit perturbative threshold. Only these passages, not the entire private corpus, are claimed read. No private title, passage, or hash was sent in a network request. For the additional extension we also read the actual `TR-item-047-astra.tex`, lines 291–440, on exact identity certificates and residual-aware sum-of-squares bounds. Its residual estimate assumes a proved compact box. That estimate is not applied to the unbounded amplitude ray; (9.3) is an exact identity.

For a compact section interval I, the private theorem requires a total first return before a finite T_*, a compact flow tube, uniform transversality $\tau > 0$, an exhaustive finite interval cover, and exactly one selected oriented intersection per counted orbit. Its interval Newton boxes require $0 \notin D'(J)$ and an interior Newton inclusion. These certify simple return zeros only. Failure at a multiple zero, tangency, or divergent return time means "not certified," not "no cycle." In the present system the source proves the one-positive- intersection property, but compact coverage and remainder bounds must still be supplied separately for a numerical certificate.

The event-corrected derivative for a section $h(z) = 0$ is

$$T'(s) = -\frac{\nabla h \cdot V}{\nabla h \cdot X}, \qquad P'(s) = D\sigma \left[I - \frac{X \nabla h^T}{\nabla h \cdot X} \right] V, \tag{2.1}$$

where X is the vector field and V the fixed-time initial-data variation at the return. The denominator needs a uniform nonzero bound on the event enclosure. A fixed-time variational calculation without this correction is not a return derivative.

For $\varepsilon^{-k} D = M + \varepsilon R$ on a compact section, the private threshold is $|\varepsilon| B_0 < \min(b, d)$ and $|\varepsilon| B_1 < \kappa$, where B_0, B_1 bound R, R', b bounds endpoint signs, d excludes the complement, and κ bounds root-box derivatives. Section 8 specializes this contract to the actual source polynomial.

2.1 Public literature and version check

The read-only GitHub commit endpoint returned the same pinned commit as current `main`, with commit date October 6, 2026. This is a version observation, not a validity judgment or a claim of peer review. The public source files are compared byte-for-byte in the provenance script; its frozen output records the exact retrieval time and result.

Rychkov's 1975 result concerns the *odd* quintic primitive [2]; it does not dispose of arbitrary x^2, x^4 terms. Li and Lu's primary article concerns nondegenerate slow–fast cycles and slow divergence integrals [3]. Its limiting-regime hypotheses must not be discarded. The August 20, 2026 primary preprint of Chen, Li, Zhang, and Zhang explicitly distinguishes that result from the then-open unrestricted degree-five question in its Introduction [4]. Its own theorem concerns piecewise-linear systems and is not a polynomial quintic theorem. The September OpenAI claim postdates that discussion; the two statements are not contemporaneous contradiction evidence.

The source also cites the quartic uniqueness theorem of Li and Llibre, and degree-six-and-higher counterexamples. Direct DOI opens for the quartic and degree-six papers failed in this session, so their full proofs are not claimed independently checked. They are historical context attributed to the source and to the accessible primary Introduction just cited. Classical Schwarzian identities are derived where used below; an author's account is available in Ovsienko–Tabachnikov [5]. No broad literature search can certify historical novelty of the elementary local results here.

3 Half-Arcs and the Full Matching Domain

3.1 Coordinates and exact transport

Translate y to remove $F(0)$. On either open half-plane y is strictly monotone in physical time. The substitution $u = x^2/2$ gives

$$u_y = \phi(u) - y, \qquad \phi_\pm(u) = F(\pm\sqrt{2u}). \tag{3.1}$$

For now take any $\phi \in C^\infty((0,\infty); \mathbb{R})$, peak $t > 0$, and $0 < h < t$. The solution through $(\phi(t), t)$ has exactly two level-h endpoints $y_< < \phi(h) < y_>$. Indeed $u_{yy} = \phi'(u) u_y - 1$ makes every critical point

a strict maximum; boundedness of ϕ on $[h,t]$ forces both descending branches to reach h at finite y. Set

$$r = (y_> - y_<)/2, \quad M = (y_> + y_<)/2 - \phi(h), \quad v = M_r.$$

The initial-data variation $\xi = \partial_t u$ satisfies $\xi_y = \phi'(u)\xi$, $\xi(\phi(t);t) = 1$, hence $\xi > 0$. Implicit endpoint differentiation yields

$$(y_<)_t = -\xi(y_<)/(r-M) < 0, \qquad (y_>)_t = \xi(y_>)/(r+M) > 0, \tag{3.2}$$

$$r_h|_t = -r/(r^2 - M^2), \qquad M_h|_t = M/(r^2-M^2) - \phi'(h). \tag{3.3}$$

Thus $|M| < r$, $|v| < 1$, $r_t > 0$, and $r^2 \geq 2(t-h)$. The nondegenerate maximum gives $r \downarrow 0$ as $t \downarrow h$, so for fixed h the width ranges over $(0,\infty)$. These statements are also valid for real $h < t$ when ϕ is smooth on the entire real line.

Put $q_0 = r^2 - M^2$, $\alpha = r/q_0$, and $D = \partial_h - \alpha\partial_r$. Direct differentiation, with the commutator $D(f_r) = (Df)_r + \alpha_r f_r$, gives

$$DM = M/q_0 - k(h), \qquad Dv = -2Mr(1-v^2)/q_0^2, \qquad k = \phi'. \tag{3.4}$$

The commutator term is essential. With $\theta = \operatorname{arctanh}(M/r)$ and $L = \operatorname{arctanh} v$, the outward characteristic derivative is $-D/\alpha$, so

$$h' = -r\operatorname{sech}^2\theta, \quad \theta' = k(h) - \frac{\sinh(2\theta)}{r}, \quad L' = \frac{\sinh(2\theta)}{r}. \tag{3.5}$$

Signed auxiliary-width scaling of the profile, not a negative physical width, gives smooth remainders in $M = k(h)r^2/3 + O(r^3)$ and $v = 2k(h)r/3 + O(r^2)$. Consequently the integral for L from width zero converges.

3.2 Why height zero does not require smoothness of the profile

The polynomial F is smooth, but $\phi_\pm$ may contain $u^{1/2}$. Their derivatives need not exist at zero. Suppose only that $\phi \in C^\infty((0,\infty)) \cap C([0,\infty))$, $\phi(0) = 0$. For each peak, endpoints converge to $a(t) \leq 0 \leq b(t)$. Writing $B = \sup_{[0,t]} |\phi|$, integration of $u_y \leq B - y$ and $u_y \geq -B - y$ bounds both endpoints in $[-B - \sqrt{2t}, B + \sqrt{2t}]$.

On a transverse arc, $a < 0 < b$, invert the endpoint branch:

$$y_u = \frac{1}{\phi(u) - y}.$$

Near each nonzero axis endpoint, this right side is continuous in u and smooth in y with bounded y derivatives. Variations with respect to the peak are therefore controlled down to $u = 0$ without differentiating ϕ there. Their first derivative is the initial variation times a positive exponential. Endpoint signs persist; $a_t < 0 < b_t$ persists. The transverse peak set is upward closed and open, hence empty or an open ray, whose width image is likewise an open ray. Inverting $r_t > 0$ locally shows convergence of all fixed-width derivatives as $h \downarrow 0$. These are local uniform statements on compact subsets of the transverse width domain, not estimates at its missing endpoint.

3.3 The global geometric correspondence

Let $\mathcal{I}_\pm$ be the two transverse width rays and $J = \mathcal{I}_+ \cap \mathcal{I}_-$. This is one open interval or empty. On it define

$$\Delta(r) = M_+(0,r) - M_-(0,r). \tag{3.6}$$

The source's `prop:cycle-matching` establishes a bijection between its zeros and nonconstant periodic orbit images. The argument uses all of the following facts, not simply a formal endpoint equation. Every periodic orbit has both signs of x, because $\int \dot{y}\,dt = 0$. Axis crossings are transverse, since $\dot{x} = y \neq 0$ there and the origin is the only equilibrium. Each half-map between positive and negative axis ordinates is strictly decreasing by (3.2). Their composition is increasing and cannot have a finite periodic sequence of distinct points. A periodic orbit therefore has exactly one positive and one negative intersection.

Conversely, a matched pair has common endpoints $a = M_{\pm} - r < 0 < b = M_{\pm} + r$. Reconstruction $x = \pm\sqrt{2u}$ and $dt = dy/(-x)$ gives finite travel time because $u_y(a) = -a > 0$ and $u_y(b) = -b < 0$ make the endpoint zeros simple. The two solutions join with velocity $(y, 0)$; smooth-vector-field uniqueness gives a periodic orbit. The positive ordinate $s(r) = r + M_+(0, r)$ has derivative $1 + v_+ > 0$. Local transverse flow neighborhoods identify nearby orbit images with nearby section fixed points. Hence isolated zeros correspond exactly to isolated periodic images, without a hyperbolicity assumption. Section 7 strengthens this local correspondence to an exact signed factorization.

An endpoint limiting on the equilibrium is not another periodic orbit. An interval of zeros, including its endpoints lying in J, has no isolated zeros. An isolated multiple zero is counted once. If F is even then $\phi_+ = \phi_-$ and $\Delta \equiv 0$ on J; this establishes absence of isolated periodic orbits, not a need to count each center orbit. It is not necessary to assume $J = (0, \infty)$.

4 Quadratic Fitting and the Central Model Estimate

4.1 The exact global fit asserted by the source

For $q_{\lambda,\kappa}(u) = \lambda u + \kappa u^2/2$ on $\mathbb{R}$, let $H(\lambda, \kappa, r)$ be the base-zero midpoint and put

$$P = H_\lambda, \quad Q = H_\kappa, \quad R = P_r, \quad S = Q_r, \qquad G = PS - QR.$$

Source claim. For every $r > 0$ and every $(M, v) \in (-r, r) \times (-1, 1)$, there is a unique $(\lambda, \kappa) \in \mathbb{R}^2$ with $(H, H_r) = (M, v)$. The joint map $(\lambda, \kappa, r) \mapsto (H, H_r, r)$ is a smooth diffeomorphism onto this full open data domain, and $P, Q, R, S, G > 0$. This is `thm:quadratic-fit`, not merely a small-width inversion.

For a profile variation ψ, the fixed-width midpoint variation is

$$[(M + r)K(y_>) - (M - r)K(y_<)]\delta Y = \int_{y_<}^{y_>} K(y)\psi(u(y))\,dy, \quad K(y) = e^{-\int_{y_<}^{y} \phi'(u(z))\,dz}, \tag{4.1}$$

where $Y = M + \phi(h)$ and $\delta M = \delta Y - \psi(h)$. The factor on the left is positive. Taking $\psi = u, u^2/2$ proves $P, Q > 0$. This does not yet prove the determinant sign or surjectivity.

The source parameterizes a model characteristic by peak slope z, curvature κ, and width s, with base slope j_κ and angles $\Theta_\kappa, \mathcal{L}_\kappa$. It proves $j_z > 0$, $\Theta_z > 0$, $\mathcal{L}_z > 0$, $|j - z| \leq |\kappa| s^2/2$, and that $z \mapsto \Theta_\kappa(s, z)$ is onto $\mathbb{R}$. Matching M first gives a unique slope $\lambda_*(r, M, \kappa)$. Curvature comparison, with its error equation

$$E' + (J_\kappa)_\theta E = -s(\phi''(h(s)) - \kappa)\operatorname{sech}^2\theta, \qquad E(s) = \int_0^s e^{-\int_q^s (J_\kappa)_\theta} f(q)\,dq,$$

gives a strictly positive conditional curvature derivative. Positivity of the coefficient bounds the exponential by one even if the coefficient is not integrable at zero. A finite-width quantitative comparison proves strict derivative, not merely strict monotonicity:

$$\partial_\kappa \widehat{v} = G/P > 0, \qquad \widehat{v} = H_r(\lambda_*, \kappa, r).$$

Properness is separate: if $\widehat{L} = \operatorname{arctanh} \widehat{v}$ stayed bounded as $\kappa \to +\infty$, monotone convergence would give finite interior angles almost everywhere. Two separated averages bound the drop in j, while $j(b) - j(c) = \kappa \int_b^c s \operatorname{sech}^2 \Theta \, ds$ diverges. Reflection treats $\kappa \to -\infty$.

The remaining signs follow from the model identity

$$\kappa P + \lambda = \frac{r H_r + H}{r^2 - H^2}. \tag{4.2}$$

Differentiating it and expressing $\operatorname{arctanh} H_r$ along the finite-peak characteristic proves $R > 0$ (with a separate $\kappa = 0$ calculation); $S = (G + QR)/P > 0$. These global steps are substantive premises of the later counting argument.

4.2 Transport identities and derivative conventions

For actual fitted functions $(\lambda(h, r), \kappa(h, r))$, subtract the model version of (3.4) from the actual one. The resulting linear system is

$$\begin{pmatrix} P & Q \\ R & S \end{pmatrix} \begin{pmatrix} D\lambda - \kappa \\ D\kappa \end{pmatrix} = \begin{pmatrix} \lambda - k(h) \\ 0 \end{pmatrix}.$$

Consequently

$$P\lambda_r + Q\kappa_r = 0, \quad D\kappa = -\frac{R}{G}(\lambda - k), \quad D\lambda - \kappa = \frac{S}{G}(\lambda - k). \tag{4.3}$$

The first identity differentiates the midpoint fit at fixed height. These algebraic consequences do not themselves prove any global sign for κ_r.

Source claim. The central `thm:model-inequality` says

$$C_r \le 0, \qquad C(\lambda, \kappa, r) = R/G, \quad (\lambda, \kappa) \in \mathbb{R}^2, \ r > 0, \tag{4.4}$$

where the partial derivative holds both model parameters fixed. Let $\mathcal{W} = RS_r - SR_r$. Exact quotient differentiation gives $C_r = -P\mathcal{W}/G^2$. With

$$\mathcal{D} = \kappa \partial_\lambda - \alpha \partial_r, \qquad \gamma = \frac{r^2 + H^2 + 2rHH_r}{(r^2 - H^2)^2},$$

the source derives

$$\mathcal{D}\mathcal{W} = (2\gamma + 3\alpha_r)\mathcal{W} + \gamma_r G_r - \gamma_{rr} G, \qquad \mathcal{W} = \frac{16}{45} r^3 + O(r^4). \tag{4.5}$$

At a putative zero, $R_r = \sigma_0 R$, $S_r = \sigma_0 S$, $G_r = \sigma_0 G$. The required sign is $\gamma_{rr} - \sigma_0 \gamma_r > 0$ for $\kappa > 0$. The total outward derivative is $-\mathcal{D}/\alpha$; it must not be confused with the partial derivative in (4.4).

4.3 Riccati and Schwarzian structure of the source proof

Write $\kappa = 2b > 0$, $d = \lambda$, endpoints $-m, n$, and

$$m = r - H > 0, \quad n = r + H > 0, \quad \ell = \frac{dn}{dm} = \frac{1 + H_r}{1 - H_r} > 0, \quad w = (\log \ell)_m, \quad \mathcal{S} = w_m - \tfrac{1}{2} w^2.$$

The last quantity is the Schwarzian of the endpoint correspondence. The weighted identities

$$X = \frac{\ell/m - 1/n}{1 + \ell} = \kappa P + d, \quad Z = \frac{\ell m - n}{1 + \ell} = dP + 3\kappa Q, \quad \gamma = \frac{\ell/m^2 + 1/n^2}{1 + \ell}$$

follow from (4.2) and the scaling $H(\lambda,\kappa,r)=rH(\lambda r,\kappa r^3,1)$. Define

$$D_*=(m-n)\mathcal{S}+\left[1-\frac{2n}{m}+\ell\left(\frac{2m}{n}-1\right)\right]w+\frac{m+n}{2}w^2, \tag{4.6}$$

$$w_*=\frac{2(\ell m^4-2\ell m^3n+2mn^3-n^4)}{m^2n^2(m+n)}. \tag{4.7}$$

The source's exact algebra reduces the sign at $\mathcal{W}=0$, $m\neq n$, to

$$X_mZ_{mm}-Z_mX_{mm}=\frac{\ell^2(m+n)^2D_*}{m^2n^2(1+\ell)^3}=\frac{3\kappa^2}{m_r^3}\mathcal{W}, \tag{4.8}$$

$$\gamma_{rr}-\sigma_0\gamma_r=m_r^2\frac{\ell(m+n)^2}{(1+\ell)m^2n^2}\frac{w_*-w}{m-n}. \tag{4.9}$$

The diagonal $m=n$ is dealt with before dividing by $m-n$. Likewise $X_m=\kappa R/m_r>0$ legitimizes the intermediate quotient X_{mm}/X_m; no division by w or Z_m is justified or needed.

For the actual bounded arc $u_y=du+bu^2-y$, set $V=\exp[-\int_{-m}^{y}(bu+d/2)]>0$, $U=uV$. Differentiating gives

$$U'=\tfrac{d}{2}U-yV,\quad V'=-bU-\tfrac{d}{2}V,\qquad V''=(by+d^2/4)V. \tag{4.10}$$

This regular two-dimensional linear system crosses $y=0$. The scalar U equation has a $1/y$ singularity and is only used near the nonzero endpoints. Choose a fundamental second solution; the ratio $f=U_1/U_2$ is finite and has nonzero derivative near both endpoints, with the same identity $f(n(m))=f(-m)$. After $U=\sqrt{|y|}\,\widetilde{U}$, the scalar equation is

$$\widetilde{U}''=(by+d^2/4-d/(2y)+3/(4y^2))\widetilde{U}.$$

For solutions $a'',c''=Q_0a,Q_0c$, respectively, $(a/c)''/(a/c)'=-2c'/c$ implies $\{a/c,y\}=-2Q_0$. The Schwarzian chain rule applied at the two endpoints yields

$$\mathcal{S}=2b(m+n\ell^2)+\tfrac{d^2}{2}(\ell^2-1)-d(1/m+\ell^2/n)+\tfrac{3}{2}(\ell^2/n^2-1/m^2). \tag{4.11}$$

Endpoint variation also yields

$$\ell=\frac{mV(-m)^2}{nV(n)^2},\qquad w=1/m-\ell/n+d(1+\ell)+2bmV(-m)^2\int_{-m}^{n}V^{-2}.$$

For $d<0$, $d^2\geq 4bm$, $\nu=\sqrt{d^2/4-bm}$, comparison of $V(-m+s)/V(-m)$ with $\cosh(\nu s)-d\sinh(\nu s)/(2\nu)$ gives

$$V(-m)^2\int_{-m}^{n}V^{-2}\leq\frac{1}{\nu-d/2},\qquad w\leq 1/m-\ell/n+d\ell-2\nu.$$

At $\nu=0$ the comparison is $1-ds/2$, whose inverse-square integral is $-2/d$. Thus the equality boundary is included.

The final endpoint split is indispensable. For $m\leq n$ the characteristic gives $\ell>1, w>0$; $m=n$ has $D_*>0$, and $m<n$ has $w_*<0$, hence the desired ratio in (4.9) is positive. For $m>n, \ell\geq 1$, one has $d<0$, $\mathcal{S}>0$, and a component argument on the actual m-domain proves $w\geq 0$; thus $D_*>0$. That argument does not assume the m-domain is unbounded.

For $m > n, 0 < \ell < 1$, scale by the *fixed* endpoint value n at the point and write $n = 1, m = z > 1$. The source uses

$$T = 2\ell z^3 - \ell z^2 - z + 2, \qquad A = 2\ell z^3 - 3\ell z^2 + 3z - 2 > 0.$$

At $w = w_*$, the baseline $\mathcal{S}_0 = 3(\ell^2 - z^{-2})/2$ gives $D_* = (z-1)AT/(2z^4)$, and $z^{-1} - \ell - w_* = -(z-1)T/((z+1)z^2)$. The derivative of D_* in w is positive for $w \geq w_*$. If $T \geq 0$, the Riccati bound forces $d^2 < 4bz$ under that assumption, and (4.11) strictly exceeds its baseline. If $T < 0$, one has $z > 2$, $-dz > 2$; the same bound controls a possible negative $2bz - d^2/2$ and gives $\mathcal{S} > 0$. Thus $w \geq w_*$ would force $D_* > 0$ in either subcase. At a zero, therefore $w < w_*$, giving the required positive sign.

Starting with (4.5), a first zero reached from $\mathcal{W} > 0$ would have nonpositive outward derivative; the established sign makes it positive. Reflection treats $\kappa < 0$; continuity yields the required weak sign for $\kappa = 0$. This is a reconstruction of the source's continuum argument, not a claim that our finite tests independently certify its model theorem.

5 Shape Transport and the Four Coefficient Cases

5.1 Positive-height comparisons and boundary care

At the parabolic profile the variations $u, u^2/2, u^3/6$ give midpoint/width-derivative pairs

$$(r^2/3, 2r/3), \quad (r^4/15, 4r^3/15), \quad (r^6/105, 6r^5/105).$$

Consequently the width-one Jacobian and cubic response satisfy

$$\begin{pmatrix} 1/3 & 1/15 \\ 2/3 & 4/15 \end{pmatrix}^{-1} \begin{pmatrix} 1/105 \\ 6/105 \end{pmatrix} = \begin{pmatrix} -1/35 \\ 2/7 \end{pmatrix}.$$

Smooth scaling proves, locally uniformly for $h > 0$,

$$\lambda = k - \phi'''r^4/35 + O(r^5), \quad \kappa = \phi'' + 2\phi'''r^2/7 + O(r^3), \quad \kappa_r = 4\phi'''r/7 + O(r^2). \tag{5.1}$$

The derivative assertion uses a smooth remainder, not formal differentiation of an unspecified big-O estimate.

Source claim. If $\phi''' > 0$ on all $u > 0$, then $\lambda < k$ and $\kappa_r > 0$ for all $h, r > 0$; signs reverse when $\phi''' < 0$. Along an outward characteristic $c(s) = \phi''(h(s))$ strictly decreases. Its comparison forcing relative to the final fitted curvature has at most one negative-to-positive change. Equal final θ and L force the comparison error to have already crossed, giving $k - \lambda > 0$. Differentiating (4.3) gives

$$D\kappa_r = \mathcal{T}\kappa_r - C_r(\lambda - k), \quad \mathcal{T} = \alpha_r + CQ/P - (C_\kappa - QC_\lambda/P)(\lambda - k).$$

Outward transport makes the independent forcing nonnegative by (4.4); the positive initial sign from (5.1) persists by an integrating factor.

The source also proves that $((\phi' - d)/u)' > 0$ on $(0, \infty)$ implies $\lambda - h\kappa < d$. For $I = \lambda - h\kappa - d$, the peak limit is $-t^2((k-d)/u)'|_t < 0$ and

$$DI = \frac{S + hR}{G}(\lambda - k).$$

At a hypothetical $I = 0$, strict profile comparison forces $\kappa > (k(h) - d)/h$, hence $\lambda > k$ and the outward derivative is negative, contradicting a first zero from below. Finally the fitted curvature lies between the minimum and maximum of actual ϕ'' on the compact traversed height interval; the bounds are strict when that curvature is nonconstant.

5.2 The unrestricted even part is retained

The reflection/time reversal $X(t) = -x(-t)$, $Y(t) = y(-t)$ replaces $F(x)$ by $F(-X)$ and preserves isolation and orbit images. It permits $f_5 \geq 0$ after the constant translation. Write

$$\phi_\pm(u) = d_0 u + b_0 u^2 \pm p(u), \quad p(u) = eu^{1/2} + cu^{3/2} + au^{5/2}, \quad a \geq 0, \tag{5.2}$$

$$d_0 = 2f_2, \quad b_0 = 4f_4, \quad e = \sqrt{2} f_1, \quad c = 2^{3/2} f_3, \quad a = 2^{5/2} f_5. \tag{5.3}$$

The even coefficients d_0, b_0 remain arbitrary throughout. At $r \in J$, fit the interior limiting data directly; the joint inverse gives finite $\lambda_\pm(0,r), \kappa_\pm(0,r)$. For the conditional midpoint function,

$$\widehat{v}_M = R/P > 0, \quad \widehat{v}_\kappa = G/P > 0, \qquad M'_\pm = \widehat{v}(r, M_\pm, \kappa_\pm).$$

Case 1: $e \geq 0, c \geq 0$**.** If $p = 0$, Δ vanishes identically. Otherwise $p > 0$ on $(0, \infty)$. For two hypothetical matched arches, their difference $w = u_+ - u_-$ satisfies $w' = [d_0 + b_0(u_+ + u_-)]w + p(u_+) + p(u_-)$ with zero endpoint values. Variation of constants contradicts its positive forcing. Thus there are no isolated cycles in either alternative.

Case 2: $e \geq 0, c < 0$**.** Here $p''' = (3e - 3cu + 15au^2)/(8u^{5/2}) > 0$. Positive-height strict curvature monotonicity passes to the nondecreasing boundary difference $q = \kappa_+ - \kappa_-$; strictness need not survive that limit. The fundamental theorem of calculus in M and κ gives

$$\Delta' = A(r)\Delta + B(r)q(r), \qquad B > 0, \tag{5.4}$$

with continuous coefficients on J. Specifically A averages $\widehat{v}_M$ on the midpoint segment at κ_+, and B averages $\widehat{v}_\kappa$ on the curvature segment at M_-. The segments stay inside the open fit domain. Multiplying by $\mu = \exp(-\int_{r_*}^r A)$ yields $(\mu\Delta)' = \mu B q$. Since a nondecreasing q has negative, zero, and positive regions in that order, $\mu\Delta$ decreases, is constant, then increases. There are at most two isolated zeros. If a constant zero segment occurs, its interior and included endpoints are nonisolated; the two strict parts then give no extra zero. A singleton minimum zero is allowed. This argument includes isolated multiple zeros directly.

Case 3: $e < 0, c \geq 0$**.** Now $p'' = (-e + 3cu + 15au^2)/(4u^{3/2}) > 0$ and diverges at zero. At a fixed transverse width the nearby peaks stay below a finite T, so $p'' \geq \delta > 0$ on $(0, T]$. The curvature-range theorem then gives $\kappa_+ \geq 2b_0 + \delta > 2b_0 - \delta \geq \kappa_-$ at height zero. At any root, $\Delta' > 0$. A differentiable function with positive derivative at every zero has at most one zero: two upward crossings would require an intervening first downward crossing. The conclusion is thus at most one cycle.

Case 4: $e < 0, c < 0$**.** The expression $up'' - p' = (-3e - 3cu + 5au^2)/(4\sqrt{u}) > 0$ yields $\lambda_+ - h\kappa_+ < d_0 < \lambda_- - h\kappa_-$ for positive height. Strict boundary separation requires more than taking limits. Along a fixed transverse peak the fitted parameters remain bounded, whereas $k_+(h) \to -\infty$ and $k_-(h) \to +\infty$. The exact DI identity therefore makes the positive-height gaps persist strictly on descending to zero: $\lambda_+(0,r) < d_0 < \lambda_-(0,r)$. At a matched midpoint, $P, Q > 0$ forces $\kappa_+ > \kappa_-$, again $\Delta' > 0$, hence at most one zero.

All cases include $a = 0$, vanishing lower odd coefficients, arbitrary even part, and all transverse amplitudes. This is the source's route to the global bound. A finite compact-section root certificate cannot replace this full-domain sign argument.

6 The Source's Sharpness Calculation

Set $F_\varepsilon = \varepsilon f$ with

$$f(x) = 4x - \frac{20}{3}x^3 + \frac{8}{5}x^5. \tag{6.1}$$

At $\varepsilon = 0$, the solution from $(0, s)$ is $(s \sin t, s \cos t)$, $s > 0$. Near $s = 1, 2$, compact smooth flow dependence and transverse crossings give a smooth first positive-axis return $P(s, \varepsilon)$ and time $\tau(s, \varepsilon)$ near 2π. On compact time segments away from the crossings the unperturbed x stays nonzero, so the nearby return really is first. The energy identity is exact:

$$\frac{P(s,\varepsilon)^2 - s^2}{2} = \varepsilon Q(s, \varepsilon), \qquad Q = -\int_0^{\tau(s,\varepsilon)} x f(x)\, dt. \tag{6.2}$$

Smoothness of this integral defines Q at $\varepsilon = 0$. Using the three sine-power integrals $\pi, 3\pi/4, 5\pi/8$ gives

$$Q(s, 0) = -\pi s^2(s^2 - 1)(s^2 - 4), \quad Q_s(1, 0) = 6\pi, \quad Q_s(2, 0) = -48\pi.$$

The implicit function theorem yields two distinct positive roots for all sufficiently small nonzero ε in a common interval. At a root, $P = s$ and $s(P_s - 1) = \varepsilon Q_s \neq 0$. For small positive ε the inner cycle repels and the outer attracts. This is a classical perturbative lower bound, not an independent exclusion of other cycles. At $\varepsilon = 0$ the annulus consists of nonisolated harmonic-oscillator orbits.

7 Exact Matching and Multiplier Interfaces

These local results require only $F \in C^\infty(\mathbb{R})$, $F(0) = 0$, and a transverse matched periodic orbit. They do not require the quadratic global fit or the model inequality. We use the already described smooth local half-arc endpoint coordinates.

Theorem 7.1 (Signed displacement factorization)**.** *Let $r_0 \in J$ be a zero of Δ and put $a_\pm(r) = M_\pm(0, r) - r$, $b_\pm(r) = M_\pm(0, r) + r$. There is a neighborhood of r_0 on which the physical first-return displacement $D_{\mathrm{ret}}(s) = P(s) - s$ obeys*

$$D_{\mathrm{ret}}(b_+(r)) = -c(r)\Delta(r), \qquad c(r) > 0, \tag{7.1}$$

with c smooth. At a root, $c(r_0) = 2/(1 - v_-(r_0))$, where $v_\pm = M_{\pm,r}(0, r)$. Thus the local zero sets, isolation, and every finite order of vanishing agree under the increasing coordinate $s = b_+(r)$.

Proof. Because $a'_- = v_- - 1 < 0$, the equation $a_-(\rho(r)) = a_+(r)$ has a unique smooth solution near r_0, with $\rho(r_0) = r_0$. The physical right half-return goes from $b_+(r)$ to $a_+(r)$, and the left half-return then goes to $b_-(\rho(r))$. Transversality and the one-positive-intersection property make their composition the nearby first positive return. Since $b_\pm = a_\pm + 2r$,

$$D_{\mathrm{ret}}(b_+(r)) = 2(\rho(r) - r).$$

On a sufficiently small convex neighborhood in J, define

$$A_a(r) = \int_0^1 a'_-(r + t(\rho(r) - r))\, dt < 0.$$

It is smooth even when $\rho = r$. The fundamental theorem of calculus gives $\Delta(r) = a_+(r) - a_-(r) = A_a(r)(\rho(r) - r)$. Thus (7.1) holds with $c = -2/A_a > 0$. At a root $A_a = v_- - 1$. Finally $b'_+ = 1 + v_+ > 0$; composition with this smooth local diffeomorphism and multiplication by a nonvanishing smooth function preserve finite vanishing order. The exact factorization also preserves arbitrary zero sets, so isolation does not require finite order or analyticity. □

Corollary 7.2 (Floquet formula and stability sign)**.** *At a matching root r_0, the nontrivial planar Floquet multiplier is*

$$\Lambda = P'(b_+(r_0)) = \frac{(1-v_+)(1+v_-)}{(1+v_+)(1-v_-)} > 0, \tag{7.2}$$

$$\Lambda - 1 = -\frac{2\Delta'(r_0)}{(1+v_+)(1-v_-)}, \tag{7.3}$$

$$\log\Lambda = 2\big(\operatorname{arctanh} v_- - \operatorname{arctanh} v_+\big) = -\int_0^T F'(x(t))\,dt. \tag{7.4}$$

The cycle is hyperbolic exactly when $\Delta'(r_0) \neq 0$. It attracts for $\Delta' > 0$ and repels for $\Delta' < 0$.

Proof. The derivative of the right half-map is $a'_+/b'_+ = (v_+ - 1)/(v_+ + 1)$; that of the left map is $b'_-/a'_- = (v_- + 1)/(v_- - 1)$. Their product gives (7.2); subtracting one gives (7.3). Equivalently differentiate (7.1) at the root and divide by b'_+. Taking logarithms gives the first equality in (7.4). For the second, Liouville's formula gives determinant $\exp(\int_0^T \operatorname{div} X\,dt)$ for the linearized flow. The tangent-flow multiplier at a periodic orbit is one, so the remaining multiplier is this determinant. Here $\operatorname{div} X = -F'(x)$, proving the assertion. Positivity means that $\Lambda \neq 1$ is precisely hyperbolicity in the one-dimensional transverse direction. The stated stability signs follow. □

Proposition 7.3 (Quantitative local margins)**.** *If, at a root, $-1 + \eta \le v_\pm \le 1 - \eta$ for some $0 < \eta \le 1$, then*

$$\frac{2|\Delta'|}{(2-\eta)^2} \le |\Lambda - 1| \le \frac{2|\Delta'|}{\eta^2}, \tag{7.5}$$

$$2|\Delta'| \le |\log\Lambda| \le \frac{2|\Delta'|}{\eta(2-\eta)}. \tag{7.6}$$

If exact and approximate slopes and the joining segments all lie in this interval, slope errors $\delta_\pm$ give

$$|\log\Lambda - \widetilde{\log\Lambda}| \le \frac{2(\delta_+ + \delta_-)}{\eta(2-\eta)}.$$

Proof. Both factors in the denominator of (7.3) lie in $[\eta, 2-\eta]$. For (7.6), integrate $(\operatorname{arctanh} v)' = (1-v^2)^{-1}$ between v_+ and v_-; it lies between 1 and $1/[\eta(2-\eta)]$ on the specified interval. Apply the same derivative bound separately to each approximation error to prove the last estimate. □

These formulas add a source-specific consistency check: an upward matching crossing must agree with a negative divergence integral. They also show exactly why $|v| < 1$ is not a numerical conditioning bound near the edges. An explicit η is needed for stable error transport. The identity alone does not certify a root location, first return, or a whole compact section's coverage.

8 Exact Perturbation Budgets for the Sharpness Family

Use the squared section coordinate $h = s^2$ and the compact interval $I = [1/2, 9/2]$. This h is an energy-coordinate parameter, not the positive base height of Sections 3–5. Compact flow dependence around the unperturbed circles gives some $\varepsilon_0 > 0$ with a smooth first positive return on all of I. Existence of this neighborhood is qualitative; its size is not computed. Normalize (6.2) as

$$\frac{P(\sqrt{h},\varepsilon)^2 - h}{2\pi\varepsilon} = W(h,\varepsilon) = M(h) + \varepsilon R(h,\varepsilon), \qquad M(h) = -h(h-1)(h-4). \tag{8.1}$$

The smooth quotient extends to $\varepsilon = 0$ through the integral formula, and smooth parameter dependence makes R smooth after shrinking the parameter interval. In this section M denotes this cubic, not a half-arc midpoint.

Lemma 8.1 (Rational root and complement margins)**.** *Let* $J_1 = [3/4, 5/4]$ *and* $J_2 = [15/4, 17/4]$. *Then*

$$M' \geq 29/16 \text{ on } J_1, \qquad M' \leq -139/16 \text{ on } J_2. \tag{8.2}$$

The root-box endpoint values, in increasing order, are $-39/64, 55/64, 165/64, -221/64$. *On* $I \setminus (\operatorname{int} J_1 \cup \operatorname{int} J_2)$, $|M| \geq 39/64$.

Proof. $M' = -3h^2 + 10h - 4$ and $M'' = 10 - 6h$. The first derivative is increasing on J_1 and decreasing on J_2; evaluation at their left endpoints gives (8.2). Direct evaluation gives the four displayed values. The critical points of M are $(5 \pm \sqrt{13})/3$. The lower one is below $1/2$; the upper one is between $5/4$ and $15/4$ and is a positive local maximum. Thus on the left complementary component the smallest $|M|$ is $39/64$, on the middle component it is $55/64$, and on the right it is $221/64$. The outer endpoint values are $-7/8, -63/8$. This proves the global complementary margin, including box boundaries. □

Theorem 8.2 (Source-specific conditional threshold)**.** *Suppose* $0 < |\varepsilon| \leq \varepsilon_0$, *the smooth first-return contract holds on all of* I, *and* (8.1) *has uniform bounds* $|R| \leq B_0$, $|R_h| \leq B_1$ *there. If*

$$|\varepsilon| B_0 < 39/64, \qquad |\varepsilon| B_1 < 29/16, \tag{8.3}$$

then exactly two cycles have their positive-axis squared intercept in I. *There is one simple root* $h_i \in \operatorname{int} J_i$ *and none elsewhere in* I. *With* $k_1 = 29/16$, $k_2 = 139/16$,

$$|h_1 - 1| \leq \frac{|\varepsilon| B_0}{k_1}, \qquad |h_2 - 4| \leq \frac{|\varepsilon| B_0}{k_2}. \tag{8.4}$$

For positive ε, *the first cycle repels, the second attracts, and their multipliers satisfy*

$$|\Lambda_i - 1| \geq 2\pi |\varepsilon| (k_i - |\varepsilon| B_1) > 0. \tag{8.5}$$

The stability directions reverse for negative ε. *No cycle outside* I *is excluded by this theorem.*

Proof. The endpoint signs of W persist by the first strict budget. The derivative has the sign of M' on each box by the second, so the intermediate value theorem and strict monotonicity give exactly one simple zero per box. The complementary margin prevents any other zero. At a zero of W, positivity of the return ordinate makes $P^2 = h$ equivalent to $P = \sqrt{h}$. The geometric one-positive-intersection correspondence prevents double counting.

At a root, $|M(h_i)| \leq |\varepsilon| B_0$; integrate the lower derivative bounds between 1 and h_1, or 4 and h_2, to obtain (8.4). Differentiate the exact identity $P(\sqrt{h}, \varepsilon)^2 - h = 2\pi\varepsilon W$ at a fixed point. The left derivative is $P_s - 1$, so $\Lambda_i - 1 = 2\pi\varepsilon W_h(h_i, \varepsilon)$. The margins in (8.2) prove (8.5) and the signs. The actual planar first return has positive derivative by Liouville's formula with the same crossing orientation; therefore the outer multiplier is between zero and one when $\varepsilon > 0$, not merely below one. □

If B_0, B_1 are positive validated numbers, the sufficient threshold is

$$|\varepsilon| < \min\left\{\varepsilon_0, \frac{39}{64 B_0}, \frac{29}{16 B_1}\right\};$$

zero bounds remove their respective constraints. This is a sharper statement than unspecified smallness because it supplies exact source-specific margins and localization constants. It is still conditional: no numerical values for B_0, B_1, ε_0 have been validated here. A C^0 remainder alone cannot exclude extra oscillatory zeros near a simple zero. Multiple roots need separate certificates and are not admitted by the derivative margins.

9 A Conditional Shape Extension

This section gives an additional conditional extension. Its exact coefficient algebra has a separate finite-check record. It does not supply an independent proof of the central model estimate. Throughout this section assume the source package: the global quadratic fit, $C_r \le 0$, smooth arc and axis dependence, the strict shape comparisons, and the full matching correspondence with their stated domains. This is an explicit mathematical hypothesis, not an assertion of independent verification.

Lemma 9.1 (Weak shape comparison, conditional)**.** *Under that package, $\phi''' \ge 0$ on $(0,\infty)$ implies $\lambda \le \phi'$ and $\kappa_r \ge 0$ for every $h, r > 0$. The inequalities reverse for $\phi''' \le 0$.*

Proof. For $\delta > 0$, perturb the profile to $\phi_\delta(u) = \phi(u) + \delta u^3/6$. Its third derivative is strictly positive, so the source's strict comparison applies. Fix $h, r > 0$. The finite arc and transverse endpoints lie in a compact positive-height neighborhood. Smooth parameter dependence, the inverse peak-to-width map, and the smooth joint quadratic inverse make the fit and its width derivative continuous in δ. Letting $\delta \downarrow 0$ gives the weak inequalities at this fixed point. No estimate uniform as $h \downarrow 0$ is used. Apply the same argument to $-\phi$ for the reversed assertion. □

Theorem 9.2 (General shape criterion, conditional)**.** *Let $F \in C^\infty(\mathbb{R})$ with $F(0) = 0$ and set $g = (\phi_+ + \phi_-)/2$, $p = (\phi_+ - \phi_-)/2$ on $(0,\infty)$. Under the source package, either condition*

$$p''' \ge |g'''| \text{ on } (0,\infty), \qquad \text{or} \qquad p''' \le -|g'''| \text{ on } (0,\infty) \tag{9.1}$$

implies at most two geometrically distinct isolated periodic orbits of (1.1) in the entire plane.

Proof. In the first case $\phi_+''' \ge 0$ and $\phi_-''' \le 0$. Lemma 9.1 makes the positive-height fitted curvature difference nondecreasing in width. For any two widths in the common transverse axis interval J, take the boundary limit at these two fixed widths. Continuity of the axis data and quadratic inverse gives a nondecreasing continuous $q = \kappa_+ - \kappa_-$ on J. The exact midpoint identity (5.4) and its positive integrating factor give at most two isolated roots by the decrease, constant, increase argument. If J is empty there are none. Every periodic orbit corresponds to an interior root, including multiple isolated roots. In the second case q is nonincreasing; apply the same argument to the negative of the weighted displacement. The argument passes inequalities to the limit, not a count of isolated roots under approximation, which would be unjustified. □

Corollary 9.3 (A sufficient nonodd septic region, conditional)**.** *For $F(x) = \sum_{j=1}^{7} f_j x^j$, suppose*

$$f_1 \ge 0, \quad f_3 \le 0, \quad f_5 > 0, \quad f_7 > 0, \qquad 64 f_6^2 \le 175 f_5 f_7. \tag{9.2}$$

There is no restriction on f_2, f_4. Under the source package, the system has at most two isolated cycles on $\mathbb{R}^2$.

The coefficient inequalities are sufficient, not a necessary classification. Their weak boundary is included within the sector $f_5, f_7 > 0$; the region is not claimed closed in the full coefficient space, and the proof below divides by the strictly positive d.

Proof. Write $e = \sqrt{2}f_1$, $c = 2\sqrt{2}f_3$, $a = 4\sqrt{2}f_5$, $d = 8\sqrt{2}f_7$, $\gamma = 8f_6$. Then $p = eu^{1/2} + cu^{3/2} + au^{5/2} + du^{7/2}$ and $g = 2f_2u + 4f_4u^2 + \gamma u^3$. For $v = \sqrt{u} > 0$ the exact identity is

$$8\sqrt{u}\,(p''' - |g'''|) = \frac{3e}{v^4} - \frac{3c}{v^2} + 105d\left(v - \frac{8|\gamma|}{35d}\right)^2 + 15a - \frac{192\gamma^2}{35d}. \tag{9.3}$$

All terms are nonnegative provided $\gamma^2 \le 175ad/64$. Since $ad = 64f_5f_7$ and $\gamma^2 = 64f_6^2$, this is exactly (9.2). It proves the shape condition on the entire positive ray. There is no unresolved polynomial residual and no promotion of sampled compact-box positivity to $u > 0$. □

The exact-identity discipline in the actual private `TR-item-047-astra.tex`, lines 291–440, requires coefficient identities to be proved exactly or their residuals to be bounded on a declared domain. Its compact-box residual formula is not a full-ray theorem. Here the residual in (9.3) is exactly zero, the denominator factors are positive for $v > 0, d > 0$, and every displayed summand has a proved nonnegative sign. Thus this particular certificate holds on the full open ray without an unproved tail cutoff or residual bound. The conditional passage from shape to cycles is still a separate analytic obligation.

Proposition 9.4 (Sharpness in the cone, conditional)**.** *Under the same package, for all sufficiently small positive ε the genuinely nonodd degree-seven primitive*

$$F_\varepsilon(x) = \varepsilon(4x - 20x^3/3 + 8x^5/5) + \varepsilon^2(x^6 + x^7) \tag{9.4}$$

has exactly two global isolated periodic orbits, both hyperbolic.

Proof. Its coefficient margin is $175f_5f_7 - 64f_6^2 = 8\varepsilon^3(35 - 8\varepsilon) > 0$ for $0 < \varepsilon < 35/8$. The conditional upper bound therefore applies in that parameter interval. Dividing the exact energy displacement by ε gives a smooth integral whose limit is still $-\pi s^2(s^2-1)(s^2-4)$: the added terms are order ε^2. The simple derivatives $6\pi, -48\pi$ at $1, 2$ give two nearby hyperbolic cycles by the same local first-return and implicit-function argument as Section 6. They are distinct because their positive intersections are distinct. For sufficiently small positive ε both statements hold; the conditional global bound excludes any others. □

The number $35/8$ is a coefficient-cone threshold, *not* a proven two-cycle existence threshold. The existence argument remains qualitative; the global exclusion remains source-dependent. The cone is restricted, so this does not restore the false degree-seven Lins–de Melo–Pugh bound for all coefficients.

For an independent coefficient box, write $b_6 = \max(|f_{6,-}|, |f_{6,+}|)$. The conditions $f_{1,-} \ge 0$, $f_{3,+} \le 0$, $f_{5,-}, f_{7,-} > 0$ and $175f_{5,-}f_{7,-} - 64b_6^2 \ge 0$ certify (9.2) for every tuple in the box, by monotonicity of the positive product and the absolute-value bound. This finite coefficient certificate controls an infinite amplitude domain via the exact identity, not via amplitude sampling. A negative box margin is inconclusive; it does not establish a third cycle.

10 Benefits, Limitations, and Research Directions

What is gained. The local displacement factorization preserves multiplicity and links source matching data directly to the physical return map. The multiplier formula supplies exact sign checks and explicit conditioning margins. The perturbation theorem gives rational endpoint, complement, derivative, and localization constants for the actual sharpness polynomial. The additional weak shape cone makes a nonodd degree-seven extension precise, including equality in the coefficient condition and exact positive-ray algebra. All local results have explicit domains and quantifiers.

What is not gained. Neither the scalar tests nor the private return machinery prove the global fit or central model inequality. The perturbative certificate does not cover missing amplitudes, the equilibrium, or the boundary of the return domain. Its simple-root requirement does not count a nonhyperbolic cycle. The conditional septic extension inherits every source-package dependency. Symbolic identities and coefficient fixtures cannot verify a continuum transport proof. No runtime, accuracy-versus-cost, historical novelty, or optimal-threshold comparison was measured.

Prioritized work. First, compute a validated first-return neighborhood and C^1 remainder budget for the sharpness family, using event correction and compact coverage. Second, test the local identity (7.4) against independently enclosed divergence integrals, with explicit η margins. Third, develop a multiple-root certificate that retains the source's geometric count convention. Fourth, study how much of the closed shape cone survives controlled nonzero residuals with a weight valid on the full ray; an unweighted compact residual bound is insufficient. Finally, independently formalize the global fitting, properness, and model-sign dependencies before promoting any conditional extension to an independent theorem.

Contribution assessment. C2 is conservative and appropriate. There are complete local proofs, exact source-specific margins, and a full conditional shape extension. Their usefulness is distinct from originality: the methods are classical, the extension depends on the source package, and historical novelty has not been established. L4 continues to express strong relevance to return-map counting, not a stronger contribution score or a global proof certificate.

A Reproducibility and Bounded Review

The report is standalone TeX; theorem environments and all mathematical statements are contained in this file. The ten actual public inputs are preserved in `evidence/public-source/`, with full-file hashes, byte equality to the pinned public raw URLs, and the current-head observation in `evidence/public-version-check.json`. The actual private path, full-file hash, and selected read ranges are recorded locally in `evidence/readscope.json`; no private material entered the public queries. The structural examples M005 and M708 were consulted for report organization, not for mathematical authority.

`author_mathtests.py` ran under the existing Python 3.10/SymPy installation: 40 exact checks passed. They comprise the symbolic multiplier subtraction, 25 rational slope fixtures, four multiplicity coordinate fixtures, endpoint values, derivative and complement margins, Sturm root count, sine integrals, the exact energy cubic, root slopes, and one explicitly hypothetical remainder-budget fixture. The fixtures are algebra tests, not numerically validated trajectories. A separate exact-check suite has 59 further passing checks of the ray identity, coefficient conversion, septic margin, leading energy integral, parabolic corrections, and finite coefficient-box regressions. Those are stored separately and do not validate the global model.

The complete assembled report received a full mathematical read; no decisive defect was found in its local arguments, source reconstruction outside the separately assigned central model, or conditional extension. The independent model review read all 913 lines of the actual `model.tex` and ran 70 exact symbolic checks, all passing. It found no decisive model-local gap, conditional on the imported arc and fitting inputs, including $P, R, G > 0$ and smooth finite-peak characteristics. In particular it checked the endpoint determinant, constrained gamma identity, Riccati and Schwarzian reductions, all endpoint sign cases, and the fixed-parameter versus characteristic derivative distinction. It did not independently audit the full arc/fitting package or certify the global quintic theorem.

A separate logical extension review found no decisive gap under the explicit source package and exact coefficient identities. Its minor qualification is incorporated: the septic coefficient region is sufficient, not necessary, and its strict $f_5, f_7 > 0$ conditions preclude ambient closure. Only the weak discriminant boundary within that positive sector is included. These bounded outcomes and their read scopes are preserved as review snapshots; they are not a claim of formal verification or historical novelty.

Native opening returned a queued result. Native compilation failed with the diagnostic "Unable to find standard directories for platform"; it is not reported as successful compilation. The separate on-disk export uses the existing installed compiler via the project exporter, with no installation. Its actual build audit, log, page count, and page-by-page visual review are retained with the final evidence. The first physical export produced 17 pages with resolved references. One overlong inline threshold was converted to displayed mathematics, and the bibliography was given its own page. The second export is the final format round. Its exact source/PDF hashes, compiler audit, and all-page visual record are part of the frozen evidence.

The frozen manifest binds final TeX/PDF bytes, read scope, source snapshots, exact tests, bounded proof reviews, and build evidence. It excludes itself and live shared catalogue, queue, progress, and other mutable project files. Source claims, proved local statements, and conditional bridges remain distinct after freezing.

References

[1] OpenAI, *Two limit cycles for quintic Liénard systems*, September 24, 2026, pinned public manuscript and ten TeX inputs. https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/two-limit-cycles-for-quintic-lienard-systems-September-24-2026.

[2] G. S. Rychkov, The maximal number of limit cycles of the system $\dot{y} = -x$, $\dot{x} = y - \sum_{i=0}^{2} a_i x^{2i+1}$ is equal to two, *Differential Equations* 11 (1975), 390–391. https://www.mathnet.ru/eng/de2400.

[3] C. Li and K. Lu, Slow divergence integral and its application to classical Liénard equations of degree 5, *Journal of Differential Equations* 257 (2014), 4437–4469. https://doi.org/10.1016/j.jde.2014.08.015.

[4] H. Chen, Z. Li, R. Zhang, and X. Zhang, *The number of limit cycles of piecewise linear Liénard systems*, arXiv:2608.19542v1, August 20, 2026, Introduction. https://arxiv.org/html/2608.19542v1.

[5] V. Ovsienko and S. Tabachnikov, What Is ... the Schwarzian Derivative?, *Notices of the AMS* 56 (2009), 34–36. https://ovsienko.perso.math.cnrs.fr/Publis/what-is.pdf.

[6] D. Muralidhara Subbarao, confidential `TR-item-015-astra.tex`, selected actual return-operator, event-derivative, and perturbative-threshold passages. Unpublished local material; provenance and read scope are recorded in the evidence. The exact-identity discipline is also informed by the selected actual `TR-item-047-astra.tex` passages, without importing its compact residual bound to an unbounded ray.